

CHAPTER 1: INTRODUCTION TO ETHICS

1. What is Ethics?

- **Ethics** = a set of values that define acceptable behaviour in society.
- Ethics helps society decide what is **right or wrong behaviour**
- Ethics acts like an “invisible guidebook” for behavior when laws don’t cover everything.
- **Computer ethics** = guidelines for morally acceptable use of computers.
- Universally wrong: lying, stealing. What’s ethical in one place may be unethical elsewhere
- Culture differences: e.g., **software piracy** might be tolerated in some countries but illegal in others.

Example: Copying Microsoft Office without a license = piracy. In Malaysia, it’s unethical and illegal, but in some countries people still see it as “normal.”

Example: A student uses ChatGPT to write an assignment. If the lecturer didn’t ban it, is it unethical? Some might say yes (dishonesty), others no (smart use of tools)

2. Morals, Ethics, and Laws – Differences

Aspect	Morals	Ethics	Laws
Based on	Religion or personal belief	Society/profession’s code of conduct	Government rules
Nature	Individual & internal	Collective & professional	Enforceable by police/courts
Example	Believing abortion is wrong	IT professional code against piracy	Abortion legalized in some countries

Contrast: A lawyer may *ethically* defend a guilty client (professional duty), though it may seem *immoral* to some.

3. Why Ethics in IT?

- IT affects **governance, work, personal life** → ethical decisions matter.
- In IT, ethical decisions often involve balancing company interests with individual rights.
- Issues include:
 - **Privacy** (employee monitoring, email usage)
 - **Data security** (protection against manipulation/leaks)
 - **Global concerns** (cross-border data, cybercrime)

Example: A company tracks employee browsing history to prevent misuse, but this raises ethical concerns about privacy.

Example : Company installs software that tracks employee keystrokes.

4. Ethics for IT Professionals

- **Who is an IT Professional?** Anyone working in IT (e.g., developers, engineers, analysts).
- **Characteristics of professionals:**
 - Advanced knowledge & creativity.
 - Independent judgment.
 - Commitment to lifelong learning.
 - Contribution to society.
- **Professional Code of Ethics:**
 - Sets out goals + rules for members.
 - Guides behaviour beyond just following laws.

Example: IEEE and ACM have professional codes guiding IT professionals to act responsibly.

Example: Doctors trust IT professionals to keep **medical data confidential**. If IT staff leak patient records, public trust is lost

5. Common Ethical Issues for IT Users

1. **Software Piracy** – illegal copying of software.
2. **Inappropriate Use of IT** – browsing, gaming, sharing offensive content at work, wasting company resources.
3. **Inappropriate Sharing of Information** – leaking confidential business or customer data.

Example: Employee shares a customer's credit card number by accident → breach of privacy.

Example : A leaked **database of credit card info** from an e-commerce website → customers may never shop online there again.

6. Certifications in IT

- Proof of skills, often boosts career & salary.
- **Vendor Certifications:** Specific to a company's products.
 - Examples: Cisco, Microsoft, AWS.
- **Industry Association Certifications:** Broader, require exams & ongoing learning.
 - Examples: (ISC)², ISACA.

Example: AWS Cloud Certification = shows ability to manage cloud services.

Example: A Cisco-certified network engineer might be hired faster because certification reassures employers of basic competence

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CHAPTER 2: COMPUTER AND INTERNET CRIME

1. What is Cybercrime?

- Illegal activity using the Internet.
- A computer can be:
 - **Weapon** (used to launch attack),
 - **Target** (hacked),
 - Or **both**.
- Two types:
 1. **Malicious harm** (e.g., stealing data)
 2. **Accidental harm** (e.g., employee accidentally spreading virus)

Example: Creating a virus vs. an employee “accidentally” introducing malware.

Example: Clicking a phishing email link → accidentally exposes company network to hackers.

2. Computer User Expectations

- Technology allows **faster, cheaper data storage**.
- More data collected, easier to track users
- Audit trails = record every modification.
- Problems:
 - Users share login credentials → unauthorized access.
 - Increasing complexity → more vulnerabilities.
 - Rapid tech changes → harder to keep secure.

Example: Sharing your office Wi-Fi password with a friend can risk company data leaks.

Example: Using **fingerprint login** is convenient, but if stolen, you can’t “reset” your fingerprint like a password

3. Reliance on Commercial Software

- Weaknesses in software = **exploits**.
- Companies release **patches** to fix flaws, but if users don’t install → risk of breach.
- Most cyberattacks happen because people don’t install patches in time.

👉 **Example:** WannaCry ransomware spread because people didn’t install Windows updates.

4. Types of Security Attacks

A. Passive Attacks (stealing info, no damage)

- **Eavesdropping** (listening to calls/emails).
- **Traffic monitoring** (observing communication patterns).
- Hard to detect.

B. Active Attacks (tampering/disrupting)

1. **Masquerade** – pretending to be someone else.
2. **Replay** – resending captured messages.
3. **Message alteration** – changing/delaying messages.
4. **Denial of Service (DoS/DDoS)** – overwhelming a system with fake requests.

Contrast: Passive attack = silent spying. Active attack = visible damage.

Passive attacks are like “silent thieves,” active attacks are like “burglars smashing windows.”

Passive: Hacker listens to your Wi-Fi traffic.

Active: Hacker floods your Wi-Fi router so it crashes.

5. Common Malware & Threats

- **Virus** – attaches to files, spreads when opened.
- **Worm** – spreads automatically (no user needed).
- **Trojan Horse** – disguised as legit software, cannot self-replicate.
- **Botnet** – network of infected computers controlled by hacker.
- **Rootkit** – hides attacker’s presence.
- **Spam** – unwanted messages (email, SMS, social media).
- **Phishing** – tricking users to reveal sensitive info (passwords, credit cards).

Example: A fake “bank email” asking you to click a link is phishing.

Example: Clicking a fake “free game” download (Trojan) installs a botnet client → your PC secretly joins a DDoS attack

6. Hackers vs Crackers

- **Ethical Hackers:** Test systems legally, help improve security (e.g., CEH certified).
- **Crackers:** Malicious hackers who break systems for personal gain.

Contrast: Ethical hackers = white hats. Crackers = black hats.

Example: Facebook pays **bug bounty hunters** (ethical hackers) who report security flaws

7. Malicious Insiders

- Employees or ex-staff abusing access.
- Often collude with outsiders for fraud.

Example: Bank employee leaking customer account details.

Example: IT admin selling confidential student data from a university system

8. Cybercriminals Categories

1. **Cybercriminals** – financially motivated (stealing credit card data, funds).
2. **Hactivists** – politically/socially motivated.
3. **Cyberterrorists** – aim to cause large-scale harm (e.g., hacking air traffic systems).

Motives matter — the same hacking skill can be used for money, protest, or destruction.

Example:

- Cybercriminal: Stealing credit cards.
- Hactivist: Defacing a government website to protest policy.
- Cyberterrorist: Attacking air traffic systems to cause chaos.

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CHAPTER 6: RISK ASSESSMENT AND TRUSTWORTHY COMPUTING

1. General Security Risk Assessment

- **Definition:** A systematic way to identify risks to an organization's IT systems (hardware, software, databases, networks).
- **Why Important:** Helps decide how much money and time should be invested in protecting the organization.

Key Concepts:

- **Assets:** Anything valuable (e.g., servers, data, applications).
- **Failure Case:** Any harmful event (e.g., virus infection, DDoS attack).
- **Risk Formula:**

$$\text{Risk Level} = \text{Probability of Threat} \times \text{Impact of Threat}$$

Example:

- If a hospital's patient database (asset) is hit by ransomware (failure case), the **impact** is high (lives at risk, legal issues). Even if probability is moderate, the risk is **critical**.

2. Trustworthy Computing

- **Definition:** An approach ensuring technology is **secure, private, reliable, and ethical**.
- **Goal:** Build confidence in computing for customers and businesses.

Core Principles:

1. **Security** – Systems resist attacks.
Example: Firewalls and antivirus software.
2. **Privacy** – Users control their personal data.
Example: GDPR compliance in Europe.
3. **Reliability** – Systems work consistently.
Example: Online banking systems available 24/7.
4. **Business Integrity** – Companies act ethically.
Example: Transparent billing by cloud providers.

3. CIA Triad

The foundation of cybersecurity:

- **Confidentiality** → Only authorized people can access data.
Example: Password-protected medical records.
- **Integrity** → Data cannot be altered improperly.
Example: Digital signatures to prevent tampering.
- **Availability** → Systems must be accessible when needed.
Example: E-commerce sites working during Black Friday.

4. IT Security Policy

- **Definition:** A written plan on how a company protects its IT assets.
- **Examples of Policies:**
 - Password policies (e.g., must change every 90 days).
 - Remote access rules (VPN required).
 - Data backup and disaster recovery procedures.

5. International Standards (ISO 27001)

- **What is it?** A global standard for **Information Security Management Systems (ISMS)**.
- **Covers 14 areas**, such as:
 - Access control (who can access what).
 - Cryptography (data encryption).
 - Human resource security (training employees).
 - Compliance with laws.

Example: A company certified under ISO 27001 can prove to customers that their data is handled securely.

6. NIST Cybersecurity Framework (CSF)

- **Created by:** U.S. National Institute of Standards and Technology.
- **Purpose:** Helps organizations **identify, protect, detect, respond, and recover** from cyber threats.

Three Main Parts:

1. **CSF Core:** 5 Functions

- Identify → Know your assets and risks.
- Protect → Safeguards (firewalls, training).
- Detect → Find security incidents early.
- Respond → Take action to contain impact.
- Recover → Restore systems and operations.

2. **Organizational Profiles:**

- *Current Profile* → Where we are now.
- *Target Profile* → Where we want to be.

3. **CSF Tiers:** Levels of implementation (Tier 1 = low, Tier 4 = adaptive & advanced).

Example: A bank might be at Tier 3 (repeatable, well-documented security) and aim for Tier 4.

CHAPTER 7: SOFTWARE DEVELOPMENT

1. Software Development Methodology

- A structured way to create software **efficiently and with quality**.
- **Popular Methods:**
 - **Waterfall:** Step-by-step, no going back. Good for projects with fixed requirements.
Example: Government tax software system.
 - **Agile:** Flexible, iterative, feedback-driven. Good for changing requirements.
Example: Mobile app updates every 2 weeks.

Cost of Fixing Errors:

- Fixing a bug early (design stage) is **100x cheaper** than fixing it after release.
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2. Software Testing

Ensures the software works as expected.

Types of Testing:

- **Manual Testing:** Human testers simulate real usage.
- **Automation Testing:** Tools/scripts automatically test functionality.
- **Dynamic Testing:** Runs the program to check behavior.
 - **Black-box:** Tester doesn't know the code, only checks inputs & outputs.
 - **White-box:** Tester knows code structure and checks all logic paths.

Example:

- Black-box → Testing login form with valid/invalid inputs.
 - White-box → Checking that every line of a function runs at least once.
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3. Quality Assurance (QA) & Standards

- **QA Definition:** Ensures software is *"fit for use."*
- **PDCA Cycle (Deming Cycle):**
 1. Plan → Define objectives.
 2. Do → Implement.
 3. Check → Monitor results.

4. Act → Improve process.

Example: If user complaints rise, QA reviews the process and updates testing.

4. Software Quality & Assurance Techniques

- **Key Attributes of Quality Software:**
 - **Scalability:** Runs on many devices.
 - **Usability:** Easy for all users.
 - **Reusability:** Code components reused in other projects.
 - **Correctness:** Meets specifications.
 - **Maintainability:** Easy to fix and update.

Example: A mobile payment app should be usable by all (simple UI), scalable (works on Android & iOS), and maintainable (bugs fixed quickly).

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CHAPTER 8: INTELLECTUAL PROPERTY (IP)

1. Introduction to IP

- **Definition:** Creations of the mind (inventions, art, designs, brand names).
 - **Why Important:** Can be more valuable than physical assets. Protects innovation and creates competitive advantage.
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2. Intellectual Property Rights (IPR)

- Rights given to creators to protect and benefit from their work.
 - Prevents unauthorized use.
 - Examples: patents, trademarks, copyrights, trade secrets.
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3. Types of Intellectual Property

a) Patents

- Protect inventions.
- In Malaysia → Governed by **Patents Act 1983**.
- Requirements: New, inventive, and industrially applicable.
- Valid for 15 years from grant.

Example: Apple patents iPhone touchscreen technology.

b) Trademarks

- Distinguish goods/services from competitors.
- Governed by **Trademarks Act 2019**.
- **Types of Trademarks (distinctiveness spectrum):**
 - Generic (weakest: "Milk")
 - Descriptive ("Fast Cleaners")
 - Suggestive ("Netflix")
 - Arbitrary ("Apple" for computers)
 - Fanciful/Coined ("Kodak")

Example: Nike "Swoosh" logo.

c) Copyrights

- Protects artistic/literary works.
- In Malaysia → **Copyright Act 1987**, updated 2012 & 2022.
- Lasts **life of creator + 50 years**.
- Based on international conventions (Berne Convention).

Example: A song, novel, or software code.

d) Trade Secrets

- Confidential info giving business advantage (e.g., Coca-Cola recipe).
 - In Malaysia → Not a specific law, but protected under **contracts and common law**.
 - Protection via NDAs, access control, employee agreements.
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4. Issues in IP

- **Plagiarism:** Copying without credit. Common in schools, writing, software.
- **Reverse Engineering:** Disassembling products to copy designs. Legal in some contexts but ethically debated.
- **Open Source vs Closed Source:** Open = free to use & modify (Linux). Closed = proprietary (Windows).
- **Competitive Intelligence (CI):** Gathering patent/market data to monitor competitors.
- **Trademark Infringement:** Using another's mark illegally (fake Adidas shoes).
- **Cybersquatting:** Registering domain names of famous brands for profit or fraud.